Script Notes and Developer Log

C.J. Tinant

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Contents

0.1 Overview

This document collects implementation notes, design decisions, and development comments across scripts used in the regional skew estimation project. It supports ongoing improvements and serves as a lightweight developer log.

Use this log to track:

- Known issues or limitations
- Planned improvements (`TODO`)
- Script-specific decisions or reminders
- Ideas for refactoring or cleanup

0.2 Project Notes (General)

0.3 TODO (General)

0.4 Future Ideas

- Add `docs/variable_scaffold.csv` after covariate extraction to distinguish between exploratory output and final cleaned results.
- Adopt a targets-based workflow {targets} after covariate extraction.

0.5 Notes by Script

0.5.1 `01a_download_ecoregions.R`

- Original metadata & layer files for each level are downloaded for reference.
- Data sources are EPA/CEC shapefiles hosted via AWS links.
- This script assumes internet access and local write permissions.

0.5.2 01b_download_USGS_gage_data.R

- Requires internet access to download data from USGS NWIS.
- Bounding box grid helps avoid request size limitations in NWIS queries.
- Uses a batch download approach for peak flow data retrieval.
- Great Plains extent is defined using EPA Level 1 Ecoregion shapefiles.

0.5.3 01d_download_us_peakflow_data.R

- The script inputs data from `01c_download_usgs_gage_metadata.R`
- The script does not apply filtering on regulation, dam failure, or record length.
- Filtering occurs in the subsequent script – `01e_filter_usgs_peakflow_data.R`

0.5.4 01e_filter_usgs_peakflow_data.R

- Bulletin 17C explicitly recommends using the Multiple Grubbs-Beck Test (MGBT) to identify low-end outliers, e.g. zero flows, and near-zero flows like 0.01, 0.02, 0.05, etc. MGBT() flags observations below a calculated threshold discharge.
- The inclusion of outliers (note in FFA there are only low-end outliers) distorts the estimation of the log-Pearson Type III distribution, biasing skewness, mean, and standard deviation. PILFs tend to skew the lower tail, by underestimating the sample skewness and producing unrealistic estimates for rare floods.
- Peak codes are comma-separated and must be parsed for reliable flagging
- Skewness is calculated as type = 1, using a method of moments estimator which follows the guidances under Bulletin 17C
- Install peakfq directly from USGS GitLab `remotes::install_gitlab("water/peakfqr", host = "code.usgs.gov")`

0.5.5 01f_download_nhdplus_v2.R

- The code retrieves flowlines and catchments intersecting the buffered AOI
- `get_nhdplus()` loops through 144 tiles that intersect AOI.
  - It fetches data chunk-by-chunk and begins stitching them together.
  - Midway or After Completion: Invalid Geometry Detected
  - The function detects one or more invalid geometries (e.g., self-intersecting polygons or degenerate line segments).
  - It triggers a geometry repair step internally – sf / s2 Geometry Repair Mode Kicks In. This temporarily activates s2 geometry engine to fix topological errors, which is common in large hydrologic datasets. You see:
    * Found invalid geometry, attempting to fix.
    * Spherical geometry (s2) switched on
    * Spherical geometry (s2) switched off

    - * Tiles Are downloaded again. The function starts over, reloading the same set of tiles (tiles 1–144) with geometry corrections in place.
  - This is expected behavior in nhdplusTools when:
    - * An invalid feature is encountered (common with complex catchments or clipped geometries), and get_nhdplus() must retry with geometry repair enabled.

0.5.6 01g_download_nhdplus_hr.R

Key Features of the script: - Idempotent: skips previously downloaded regions - Geometry-safe: cast, buffer, validate for AOIs - Fuzzy retry logic: name correction and tile fallback - Transparent logging: CSV-based log with timestamped status - Interactive QA: mapview preview of results

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0.5.7 01m_download_nlcd_2016.R

The NLCD raster was manually downloaded. Using {FdData} All tile requests timed out at ~30 seconds. Received over 100–200 MB per tile, but not enough to complete the download. Then FedData::get_nlcd(),tried to crop those partial files, which resulted in a crash.

The script inputs the following files and projects the raster to the vector CRS.

- a raster file: a SpatRaster (e.g., the full NLCD)
- a vector file: a SpatVector (e.g., Great Plains boundary)

Next, the script applies the following functions:

- crop() – Trims the raster extent down to the bounding box of the vector geometry (shape). Keeps all raster cells that intersect the box — even if they fall outside the exact shape.
- mask() – Replaces all raster cells outside the exact shape with NA. Keeps only values within the actual polygon boundary.

Added later: Paying attention to raster type is an important safeguard. I ran into a time-consuming issue when summarizing raster values for a categorical classification. The issue was that the raster was being automatically saved as floating point, FLT4S (see below).

The fix was to redownload the raster and ensure the raster is saved as INT1U.

The range of NLCD raster integer values should be between 11 to 95, which represents different types of land cover (see NLCD LUT in /docs). Thus, saving as 'INT1U', an unsigned 8-bit integer raster, enforces that values stay integers, and it signals downstream tools not to expect fractions.

The appropriate format for saving an 'INT1U' raster is as follows: INT = integer 1 byte = 8 bits per pixel = e.g. 2^8 = 256, so values range from 0–255 U = unsigned (can't store negatives)

Other common terra datatypes are: INT1U = unsigned 8-bit integer INT2S = signed 16-bit integer (–32,768 to 32,767) INT2U = unsigned 16-bit integer (0 to 65,535) FLT4S = 32-bit float FLT8S = 64-bit float (double precision)

0.5.8 01n_download_ned.R

2025-10-07 Fixed issue with mismatched tile grids (different resolutions/origins) that caused an issue with mosiacing.

docs/README version

Source: Elevation fetched via elevatr at zoom 10 (Web-Mercator).

Nominal resolution (z=10) at macrozone latitudes (~35–50°N): ~100–120 m/px.

Warped tile resolution (EPSG:4326) chosen automatically during per-tile reprojection: ~60–72 m/px (Y) with varying grid origins (see data/meta/2025-10-07_ned_tiles_resolution_z10.csv).

Analysis grid: Reprojected to EPSG:5070 and resampled to a fixed 90 m cell size prior to mosaicking and summaries.

Short Version The underlying cause of the issue was each of the 36 tiles came back at the same zoom (z = 10) but were each reprojected & clipped independently. During the per-tile warp from the service's native grid to EPSG:4326, GDAL/terra auto-chooses a target cellsize and grid origin from each tile's extent. Because each tile sits at a different latitude and has a slightly different bbox, the result is tiles with discrete degree-resolutions and origins: the range of the resulting degree-resolutions was between [0.000536, 0.000648] degrees (see results at data/meta/2025-10-07_ned_tiles_resolution_z10.csv).

Long Version The reason for the issue is the source elevation tiles are served on a Web-Mercator tile pyramid at a fixed zoom z (used z = 10). The tiles were downloaded using EPSG:4326 with clip="locations". Each tile is warped separately to lon/lat and then trimmed to your polygon. In that warp, the target resolution (in degrees) is computed automatically to roughly preserve input pixel density. Because meters/px in Web-Mercator varies with latitude, the degrees/px picked for each tile ends up slightly different.

Each tile also gets its own grid origin, so even tiles with the same res_x/res_y won't line up unless you snap them. Results explained:

- `res_y` values were ~0.000536–0.000648 deg, which is translate to ~60–72 m north–south per pixel (1-deg lat   111,132 m).
- `res_x` values also vary by cos(latitude). At 40 deg N a 0.000613 deg longitude pixel is   0.000613 × 111,320 × cos(40 deg)   52 m.

For *Web-Mercator* ground resolution at Great Plains latitudes (~35–50 deg N), z = 10 is ~100–120 m/px (not 60–72 m per-tile warp for EPSG:4326). GDAL/terra auto-picks a denser degree grid for each tile (oversampling), so the derived meters/px look finer than the z = 10 nominal.

*Nominal (Web-Mercator) resolution at z=10:

$$r(z, \phi) = 156543.0339 \cdot \cos(\phi)/2^z \text{ m/px}$$

Examples: 40 deg N is ~116.7 m; 45 deg N is ~108.1 m; 49 deg N is ~100.3 m.

Tiles were fetched at z = 10, then reprojected & clipped to lon/lat independently. In that warp, each tile got its own degree cellsize and origin to "preserve detail," often oversampling to ~60–72 m north–south.

Cellsize decision for macrozones

For macrozones: I used a z = 10 request and a 90 m cell size, because the 60–70 m in EPSG:4326 is oversampled, which looks sharper but adds no new information (false precision from per-tile oversampling). Note, for finer scale (~60–70 m) output, use a z = 11 request, which will result in heavier (greater) download density (time) and a greater number of timeouts.

Slope was calculated using (Fleming & Hoffer / Ritter algorithms), which use only the four cardinal directions (rook's case) to produces smoother, more generalized slope surfaces, which better matches subtle terrain transitions, especially in agricultural, prairie, or floodplain contexts.

When clipping I used `expand = 1000` for buffer to help prevent clipping artifacts near edges

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0.5.9 01o_download_modis_ndvi_2016.R

NDVI, or Normalized Difference Vegetation Index is a measure of photosynthetic activity occurring within a pixel, which is commonly used to assess the health and density of vegetation. NDVI ranges from [-1, +1], with healthy vegetation typically between [0.6, 0.9]. Higher NDVI values indicate healthier, more vigorous vegetation.

- NDVI = (NIR - Red) / (NIR + Red)

NDVI for the Great Plains ecoregion is calculated from MODIS, or Moderate Resolution Imaging Spectroradiometer data on the NASA Terra satellite. The relevant data, 250m resolution, 16-day composites, are labeled as `MOD13Q1`.

- MOD = MODIS sensor on Terra satellite
- 13 = Product suite 13: Vegetation Indices
- Q1 = 16-day temporal composite
- 061 = Collection 6.1 (latest operational version)

MOD13Q1 tiles contain multiple subdatasets (SDS): - NDVI — Normalized Difference Vegetation Index - EVI — Enhanced Vegetation Index - VI Quality — Bit-packed QA layer - Reflectance — Red, NIR, Blue bands used for index calculation - Day of Year — Date of composite

MODIS data are projected to a sinusoidal projection to preserve the relative size of land areas on the Earth's surface. The sinusoidal projection is an equal-area projection. MODIS tiles are 10-degrees by 10-degrees at the equator, and are named using Cartesian coordinates such that a tile named like h10v05 indicates the tile is the tenth tile from the origin in the horizontal direction and the fifth tile in the vertical direction from the origin (see https://modis-land.gsfc.nasa.gov/MODLAND_grid.html).

NDVI was chosen over EVI or Enhanced vegetation index for the Great Plains ecoregion. EVI is an index similar to NDVI that reduces the effects of: canopy background, aerosols (blue-band correction), and soil brightness, which often improves performance in dense forests. NDVI was selected for this project for the following reasons:

- Moderate to sparse vegetation: prairie, cropland, rangeland.
- Low cloud/aerosol interference: no need for blue-band correction.
- Wide use in agriculture and rangeland applications, such as crop condition, forage productivity, and drought/grazing assessment (VegDRI, USDM, NDMC).
- Fewer assumptions and post-processing corrections.

0.5.10 02a_merge_nhdplus_hr_flowlines.R

Type coercion is the process of converting a value or object from one data type to another. This can occur either automatically (implicitly) or intentionally (explicitly). Explicitly fixes issues using a file-reading loop with type coercion and error logging. The loop fixes the following issues: - Only coerce if possible, using a safe test like is.integerish(). - Fallback to numeric when needed. - Applies coercion one column at a time - Skips columns that cause errors (and logs them) - Prevents across() from failing all at once - Wrap the across() call in logic that filters to only existing columns. - Fine-grained control: logs problems per column, per file - Non-blocking: does not interrupt the whole region's read if a single column fails - Quiet warnings: keeps logs readable but still lets you know what happened - coerce fdate to character safely in all files. Character is the most flexible, readable, and safest for uncertain timestamp formats.

0.5.11 02b__merge__nhdplus__hr__catchments.R

0.5.12 02c__validate__spatial__metadata.R

0.5.13 02d__extract__metadata__from__script__names.R

0.5.14 02e__gp__parts__qc__and__outline.R

The original extent for the Great Plains Level 1 Ecoregion contains small polygons along the Texas-Louisiana Coastal Plain and the disjunct Texas Blackland Prairies.

The Texas Blackland Prairies region forms a disjunct ecological region, distinguished from surrounding regions by fine-textured, clayey soils and predominantly prairie potential natural vegetation. The predominance of Vertisols in this area is related to soil formation in Cretaceous shale, chalk, and marl parent materials. Unlike tallgrass prairie soils that are mostly Mollisols in states to the north, this region contains Vertisols, Alfisols, and Mollisols.

0.5.15 utils/process__geometries.R

Centroid locations, which the function extracts are useful for map labeling, visualization, and spatial summaries. The function applies strict checks for geometry validity and centroid coordinate extraction to avoid errors during plotting or labeling. The safeguards include checks for NA geometries, checks for empty or NULL centroids, explicit verification of coordinate matrix structure before extraction. Some addional details about process_geometries():

Geometry Validation: `st_make_valid()` is used to correct any invalid geometries within the spatial data frame. The function identifies indices of valid geometries to ensure that centroids and subsequent operations are only applied to them.

Centroid Calculation: `st_centroid()` is applied conditionally only to valid geometries using ifelse(). If a geometry is invalid, NA is used as a fallback. st_centroid() is applied only to the valid parts of the geometry column. Centroids are stored in a list to handle any type of geometry being returned from st_centroid(). Each geometry is processed individually within a loop to allow more control over handling each item and better debugging capabilities if errors occur. !is_empty(centroid) checks if the centroid is not empty. The function also ensures that st_coordinates(centroid) actually returns a non-empty data frame before trying to access its elements.

Conditional Coordinates Extraction: The x and y coordinates are extracted from the centroid. If the centroid is NA (because the geometry was invalid), the coordinate fields are set to NA. Coordinates are extracted only if there are valid centroids. This is safeguarded by checking if there are valid indices before attempting to extract coordinates. `text_x` and `text_y` are initialized with NA_real_ to ensure that the type consistency is maintained for cases where centroids might not be computable. Before extracting coordinates, the function checks if the centroid is not NA and contains rows. Then it ensures that the coordinates can be indexed properly, and has the required number of columns (at least two, for x and y coordinates).

Additional checks: The function has additional checks for NAs and data structure validity. The function checks if centroid is not NA and not empty. Then, if coordinates are derived, the function ensures it is not NA and has the necessary rows and columns. The function avoids coercion errors by ensuring each part of the conditional is valid before evaluating the next part, this prevents logical operations on possibly undefined or inappropriate data types. The function logic is structured to progressively verify conditions before accessing potentially problematic attributes like the number of rows or columns. The function ensures that coords is not null before proceeding to check its dimensions. This prevents logical errors when coords might be an unexpected type or structure. Rhe function more reliably ensures the data structure is correct before attempting to access its elements by using is.null along with checks for the number of rows and columns in `coords`

0.5.16 03a_update_covariate_metadata.R

Notes on project data structure: - Project datasets consist of feature data, vector data, and raster data stored in `~/data/processed/` - The response variable is station skew, which can be represented as vector point data. - The 63 initial explanatory variables, i.e. covariates or 'covar' in the script below, are structured by hierarchical scale (ordinal data) and domain (categorical data) - The covariate data are aggregated at each scale: Hierarchical Scale Aggregated by: 0 - Station NA 1 - Macroregional Macrozone 2 - Regional Level II Ecoregion 3 - Subregional Level III Ecoregion 4 - Local NHDPlusHD catchment

Domain Category A - Climate B - Land Cover C - Topography D - Watershed Metrics

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0.5.17 03b_station_covars.R

Station-level (Level 0) data include three location parameters: longitude, latitude and altitude, and one scale parameter: watershed area. Altitude and watershed area are reported in SI units and transformed to EPSG:5070.

The parameters for drainage area and contributing drainage area were coalesced prior to dropping gages with missing drainage area or altitude data.

0.5.18 03c_make_macrozone_lut.R

Macrozones, which represent the coarsest analysis scale for the project, are delineated along a composite gradient of moisture availability, soil development, and groundwater connectivity. These patterns are broadly reflected in three macrozones: Tallgrass Prairie, Mixed-Grass Prairie, and Shortgrass Steppe.

Ecoregion Level designations follow both North American Ecoregion nomenclature and United States (US) Ecoregion nomenclature. For example, NA_L3CODE == 9.4.1 refers to: 9 GREAT PLAINS dot 4 SOUTH CENTRAL SEMI-ARID PRAIRIES dot 1 HIGH PLAINS and correspond with US_L3CODE == 25 High Plains. The L4 designations follow the Level IV Ecoregion nomenclature (US_L4CODE), e.g. 25b the Rolling Sand Plains L4 Ecoregion of the High Plains. Macrozones are delineated at the L3 Ecoregion scale, when possible. Regions with greater variability at the L3 Ecoregion scale, e.g. the central and southern Great Plains States , i.e. Colorado, Kansas, Oklahoma, New Mexico, and Texas, exhibit a one to many cardinality at the L3 scale, and are delineated at the Level IV (L4) Ecoregion scale. Vegetative composition described in State-level L4 descriptions by Omerick and others (see Related Files below) was the primary basis for macrozone classification.

The general characteristics of the custom macrozones are as follows:

The **Tallgrass Prairie Macrozone** is located in the eastern portion of the Great Plains in the glacial till and loess plains of the Temperate Prairies Ecoregion, eastern portions of the South Central Semiarid Plains Ecoregion in the Flint Hills region, where a natural fire frequency has maintained this tallgrass remnant community. The Tallgrass Prairie macrozone is an area of higher annual precipitation (typically $>$ 850 mm), deep, fertile soils with high water retention, and dense herbaceous vegetation, and extended baseflow. Characteristic species in the macrozone include: Andropogon gerardii (Big Bluestem), Panicum virgatum (Switchgrass), Sorghastrum nutans (Indiangrass).

The **Mixed-Grass Prairie Macrozone** generally marks a transition from mesic Tallgrass Prairie in the eastern Great Plains and xeric Shortgrass Steppe in the western portion of the region, and consist of the L2 ecoregions: West Central Semi-Arid Prairies, the Northwestern Glaciated Plains, central portions of the South Central Semiarid Prairies, eastern portions of the Central Great Plains, and uplands portions of the Tamaulipas-Texas Semi-Arid Plains. The macrozone is an area of moderate precipitation (typically 500–800 mm), and high interannual precipitation variability, variable soil textures, and a combination of event-driven flow and baseflow regimes. The macrozone is characterized by a mixture of tall, mid, and short grass species, including: Schizachyrium scoparium (Little Bluestem), Bouteloua curtipendula (Sideoats Grama), Bouteloua gracilis (Blue Grama), and Stipa sp. (Needlegrass species).

The **Shortgrass-Steppe Macrozone** is located in the western portion of the Great Plains in the western portions of the South Central Semiarid Prairies, Western High Plains, Central Great Plains, and lowlands portions of the Tamaulipas-Texas Semi-Arid Plain and Southwestern Tablelands. The Shortgrass Steppe is characterized by low precipitation, sparse vegetation, shallow soils with limited infiltration, rapid surface runoff, a rapid-response, event-driven hydrologic regime, higher flood skew, driven by reduced canopy structure and limited ET buffering.

0.5.19 Related Files:

- `l3_ecoregion_descriptions.Rmd`

0.5.20 03d__make__macrozone__layer.R

The Texas-Louisiana Coastal Plain was removed from the analysis prior to developing custom macrozones. The reasons for removal include a high degree of fragmentation, a substantially different geography, e.g. coastal geography and barrier islands. Additionally, polygons less than 100 square kilometers were merged with nearest neighbors.

Macrozones are generalized following aggregation to remove polygons smaller than 1,000 sq-km or containing fewer than 30 stream gages were merged with the most ecologically similar adjacent Level III Ecoregion. The idea behind generalization is to preserve regional coherence while improving the gage count and spatial contiguity required for reliable skew estimation.

The merging process followed a hierarchical decision rule:

- Primary criterion: adjacency with a unit sharing the same Level II ecoregion classification
- Secondary criterion: ecological similarity, evaluated using Euclidean distance in a multivariate space defined by:
- Land cover composition (e.g., cropland, forest, urban fractions)
- Terrain metrics (mean and standard deviation of slope)
- Vegetation seasonality (NDVI amplitude and timing of peak greenness)

0.5.21 Zonal Summaries

Zonal summaries are used to calculate macrozone-scale patterns in climate, land cover, and topography.

The climate pattern summaries indicate broad-scale climate context, e.g. humid continental vs. semi-arid, cold hardiness diversity, and cold tolerance regimes based on minimum winter temperatures.

The scripts call utility scripts prior to calculating zonal summaries.

0.5.22 Naming convention

Short Name	Dataset	Summary Type	Variable Name
Climate Zone	Koppen-Geiger	Dominant	`kopp_geig_dom`
PHZM Zone Count	Plant Hardiness Zone	Count	`phzm_cnt`
PHZM Zone Dominant	Plant Hardiness Zone	Dominant	`phzm_dom`
Cropland Fraction	NLCD 2016 Land Cover	Fraction	`nlcd_crppst_frac`

Short Name	Dataset	Summary Type	Variable Name
Forest Fraction	NLCD 2016 Land Cover	Fraction	`nlcd_forst_frac`
Grassland Fraction	NLCD 2016 Land Cover	Fraction	`nlcd_grass_frac`
Urban Fraction	NLCD 2016 Land Cover	Fraction	`nlcd_urban_frac`
Mean Slope	NED Slope	Mean	`ned_slp_mean`
Median Slope	NED Slope	Median	`ned_slp_med`
Altitude Zone	NED Elevation	Mean	`ned_elev_mean`

0.5.23 Generalized Workflow Prior to Performing a Zonal Summary

First, run a pre-flight check with `utils/spatial/assert_inputs_ok.R` that performs the following: - Verify that rasters can be opened. - Enforce CRS and polygonal geometry. - Standardize the geometry column as `geom`. - Guarantees unique, non-NA IDs.

Second, prep and align raster and zone `utils/spatial/prep_and_align.R`. The function performs the following: - Ensures zones are valid (optional), converts to SpatVector, and reprojects zones to the raster's CRS (no raster warping). - Returns a SpatVector (polygon) in the same CRS as the raster. - Prep the raster to get zones in the raster CRS. - Optionally, crop the raster to the zones bbox and mask it to polygon shapes.

This returns a list that needs to be extracted then converted from a SpatVector to an sf object prior to {exactextractor} in `utils/spatial/rast_summ_by_class.R`

0.5.24 03e_macrozone_fix_join_key.R

Fix join key to fix an issue with Tallgrass Prairie and provide some ordinal order.

region_name	macro_id
Shortgrass-Steppe	1
Northern Mixed-Grass Prairie	2
Central Mixed-Grass Prairie	3
Southern Mixed-Grass Prairie	4
Tallgrass Prairie	5

0.5.25 03f_covar_macro_koppen_summary.R

Dominant Koppen-Geiger climate classes provides a broad-scale context on climate, e.g. humid continental vs. semi-arid.

The original Koppen-Geiger summary was a single dominant class. After an initial analysis, the zonal summary was expanded to the top n classes. Used n = 3. The original result contains 15 results. This was filtered using min_frac to minimize the number of categories and the total amount unexplained. The final result contains nine climate types across five macrozone categories with 29.9% of the variance unexplained.

Table of Results of Koppen-Geiger Climate Zonal Summary:

n-vals	max_tot_unexpl	min_frac	num_cat
15	0.212	NA	5
14	0.212	0.07	5
13	0.212	0.08	5
12	0.299	0.09	5
11	0.299	0.11	5

n-vals	max_tot_unexpl	min_frac	num_cat
10	0.299	0.135	5
09	0.299	0.14	5
08	0.441	0.15	5
07	0.441	0.25	5
06	0.441	0.35	5
05	0.618	0.37	5

Table of Results of Koppen-Geiger Climate Zonal Summary:

Macrozone	Koppen Climate Type	Description
Tallgrass Prairie	Dfa	Cold, no dry season, hot summer
Northern Mixed-Grass Prairie	Dfa	Cold, no dry season, hot summer
Northern Mixed-Grass Prairie	BSk	Arid, steppe, cold
Central Mixed-Grass Prairie	Dfa	Cold, no dry season, hot summer
Central Mixed-Grass Prairie	Cfa	Temperate, no dry season, hot summer
Southern Mixed-Grass Prairie	Cfa	Temperate, no dry season, hot summer
Southern Mixed-Grass Prairie	BSh	Arid, steppe, hot
Shortgrass-Steppe	BSk	Arid, steppe, cold

0.5.26 03g_covar_macro_phzm_summary.R

Outputs the top-3 dominant and count of phzm zones, and qa plots as a sanity check. The top-3 dominant and count of phzm zones indicates a broad-scale context on climate based on cold tolerance regimes based on minimum winter temperatures, and cold hardiness diversity. The QA plots phzm count vs area and phzm count vs latitude. The interpretation of the QA plots is as follows:

Theb phzm class count reflects both latitudinal span and region size, but the orientation of the macrozone seems to be the stronger driver, although the graph is of the centroid of the latitude.:

_ The southern Mixed Grass Prairie (round) has very few PHZM classes. The class count increases in a northerly direction. However, this is because the zones become increasingly oriented in an elongated NS direction (Tallgrass), and class count jumps. Area effect is layered in: Tallgrass Prairie has both high NS orientation, a huge area, and the highest class count (20).

0.5.27 03h_covar_make_NLCD_meta

See docs/metadata/look_up_tables/nlcd_metadata_lut.csv”) Changed classification to add Rangeland which combines Grassland, Hay/Pasture, and Shrubland. A substantial fraction of the Southern Mixed-grass Prairie is Shrubland.

0.5.28 03h__covar__macro__NLCD__meta.R

Split from 03i_covar_macro_landcover_summary to try to fix an issue with summary creating non_legend NLCD class codes. Found issue was in the way the raw raster was reprojected in R/01_download/01m_download_nlcd_2016.R. The refactored code in this script only makes a LUT.

Merged NLCD Shrub/Scrub (52) with Grassland/Herbaceous (71) and Pasture/Hay (81)) into a single Rangeland bucket. Hydrologically they behave more alike than with Forest or Crops, and it avoids a potential sparse class (Shrubland) which shows up in the southern shortgrass steppe.

0.5.29 03i__covar__macro__NLCD__summary.R

Land cover proportion summaries include the fraction of cropland, forest, rangeland, and urban land. Fraction of cropland indicates anthropogenic land use in agricultural regions associated with increased runoff, reduced infiltration, and higher ET demand. The fraction of forested land (cover) enhances interception and storage, which typically reduces peak flows and lowers skew. The fraction of rangeland moderates evapotranspiration and infiltration rates, which is especially important in mixed rangeland systems. The fraction of urban land indicates increased imperviousness, which is strongly associated with peak flow generation and positive flood skew.

0.5.30 NLCD Classes

Developed

21 *Developed, Open Space* – areas with a mixture of some constructed materials, but mostly vegetation in the form of lawn grasses. Impervious surfaces account for less than 20% of total cover. These areas most commonly include large-lot single-family housing units, parks, golf courses, and vegetation planted in developed settings for recreation, erosion control, or aesthetic purposes.

22 *Developed, Low Intensity* – areas with a mixture of constructed materials and vegetation. Impervious surfaces account for 20% to 49% percent of total cover. These areas most commonly include single-family housing units.

23 *Developed, Medium Intensity* – areas with a mixture of constructed materials and vegetation. Impervious surfaces account for 50% to 79% of the total cover. These areas most commonly include single-family housing units.

24 *Developed High Intensity* – highly developed areas where people reside or work in high numbers. Examples include apartment complexes, row houses and commercial/industrial. Impervious surfaces account for 80% to 100% of the total cover.

Forest

41 *Deciduous Forest* – areas dominated by trees generally greater than 5 meters tall, and greater than 20% of total vegetation cover. More than 75% of the tree species shed foliage simultaneously in response to seasonal change.

42 *Evergreen Forest* – areas dominated by trees generally greater than 5 meters tall, and greater than 20% of total vegetation cover. More than 75% of the tree species maintain their leaves all year. Canopy is never without green foliage.

43 *Mixed Forest* – areas dominated by trees generally greater than 5 meters tall, and greater than 20% of total vegetation cover. Neither deciduous nor evergreen species are greater than 75% of total tree cover.

Grassland

52 *Shrub/Scrub*- areas dominated by shrubs; less than 5 meters tall with shrub canopy typically greater than 20% of total vegetation. This class includes true shrubs, young trees in an early successional stage or trees stunted from environmental conditions.

71 *Grassland/Herbaceous*- areas dominated by gramanoid or herbaceous vegetation, generally greater than 80% of total vegetation. These areas are not subject to intensive management such as tilling, but can be utilized for grazing.

81 *Pasture/Hay* – areas of grasses, legumes, or grass-legume mixtures planted for livestock grazing or the production of seed or hay crops, typically on a perennial cycle. Pasture/hay vegetation accounts for greater than 20% of total vegetation.

Planted/Cultivated

82 *Cultivated Crops* — areas used for the production of annual crops, such as corn, soybeans, vegetables, tobacco, and cotton, and also perennial woody crops such as orchards and vineyards. Crop vegetation accounts for greater than 20% of total vegetation. This class also includes all land being actively tilled.

0.5.31 03j__covar__NED__summary.R

Fixed many upstream issues prior to getting the script to output reasonable results.

Step	Check
Download & Merge	All tiles successfully mosaicked
Projection	Reprojected to EPSG:5070 (NAD83 / CONUS Albers)
Extent	Clipped cleanly to updated Great Plains outline (no ocean overlap)
Slope Calculation	Derived with terra::terrain() and verified quantiles (0–83.6 deg.)
Histogram Audit	Distribution centered around ~1–5 deg., plausible for Great Plains
Outlier Handling	−133 m min elevation eliminated (buffer error fixed)
Compression/Export	Use writeRaster(…, gdal = c("TILED=YES","COMPRESS=LZW","BIGTIFF=YES"))

---

1 CHECK INTO DAGS for looking into covars.

Topography summaries include mean and median slope and mean altitude. Mean slope describes average terrain steepness, which helps distinguish rugged uplands from flatter plains. Median slope is another measure of average slope, which is more robust to outliers than the mean, and is useful for evaluating runoff tendency. Mean altitude describes the elevation regime, which influences climate, snow duration, and runoff seasonality.

1.1 Zettelkasten-style markdown

Zettelkasten-style markdown refers to using Markdown files to implement a digital version of the Zettelkasten method—a powerful personal knowledge management system developed by German sociologist Niklas Luhmann.

Zettelkasten means "slip box" in German. It's a system where you:

- Break ideas into atomic notes (each note = one concept or insight)
- Use unique IDs and bi-directional links to connect notes
- Emphasize networked thinking rather than hierarchical folders

In R or code-based workflows, Zettelkasten-style Markdown means using plain text .md files where each file is a single, focused idea, tagged and cross-linked like a mini personal wiki.

1.1.1 Example Folder Structure

```
notes/
   20250728_gam-vs-ridge.md
   20250727_skewness-viz-notes.md
   20250721_prism-uncertainty.md
   index.md  # optional index or dashboard
```

Each note might contain:

```
# GAM vs Ridge Regression for Skew Modeling

- Ridge performed best on full covariate set (low MSE)
- GAMs improved interpretability in non-linear effects (e.g. tmean_ann_C)
- TODO: test GAM + macrozone as random effect

Linked notes:
- [[20250721_prism-uncertainty]]
- [[20250727_skewness-viz-notes]]
```

1.1.2 Core Features of Zettelkasten Markdown Notes

Feature	Benefit for Researchers
Atomic Notes	Each `.md` = 1 idea = easier reuse
Links (`[[...]]`)	Creates networked ideas, not silos
Timestamps / IDs	Keeps notes organized and unique
Emergent Structure	You discover themes organically
Markdown Format	Easy to version, edit, and render

1.1.3 How It Might Help

For your hydrologic and regional skew work, this approach could help:

- Track decisions or insights across milestones
- Store modeling rationale (e.g., why you dropped ecoregions)
- Connect field notes, exploratory thoughts, and literature summaries
- Gradually build a knowledge graph of ideas and results

ChatGPT can generate a Zettelkasten-style notes/ folder with a few seed notes and a script to open/edit them easily using RStudio.

1.2 Notes

- Last updated: 2025-10-12 14:56
- Maintained by: CJ Tinant